



CONNECTMEET

A modern Zoom-style video conferencing web application for real-time online meetings

Built for the CodeMate Community

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01. PROJECT AT A GLANCE

ConnectMeet is a **Zoom-style video conferencing web application MVP** designed for real-time online meetings. The application provides a centralized interface for meeting lifecycle management, including creation, joining, and real-time collaboration tools.

The product was initially designed as a connected UI system in Design Mode before being converted into a functional MVP through Build Code Mode.

Core Capabilities

Capability	Description
Meeting Management	Instant room creation, ID-based joining, and calendar scheduling.
Real-Time Media	High-performance audio and video calling with pre-join lobby previews.
Collaboration Tools	Screen sharing, chat, emoji reactions, and raise-hand functionality.
Interaction Intel	Active speaker detection and participant pinning for focused viewing.
Host Controls	Management of participant permissions, muting, and session termination.
Reliability	Automatic reconnection logic for handling unstable network conditions.

02. PROBLEM, USERS & PRODUCT VALUE

The Problem

Remote teams and distributed groups require a unified platform for real-time video meetings that handles more than just streaming. Fragmented tools often lack integrated device permissions, connection recovery, and responsive controls required for professional collaboration.

Target Users

- **Meeting Host:** Creates rooms, manages participant settings, and controls session termination.
- **Participant:** Interacts through media toggles, screen sharing, and hand-raise gestures.
- **Meeting Organizer:** Schedules upcoming sessions and manages meeting distribution.

Product Value Proposition

ConnectMeet delivers value through a cohesive communication strategy:

03. CUSTOMER PERSONA

Primary Persona: Remote Team Member

Remote professionals need to conduct reliable meetings with simple controls for collaboration. Key goals include joining via ID, previewing media in a lobby, and switching between speaker and gallery views.

Pain Points & Solution

Common frustrations include complicated interfaces and unclear media states. ConnectMeet addresses these through a complete workflow starting from a pre-join lobby to real-time interactions with active speaker detection.

04. ARCHITECTURE & TECHNOLOGY

The application utilizes a responsive React.js architecture designed for high-performance video communication.

Technology Stack

Layer	Technology
Frontend	React.js with Component-based architecture
Real-Time Layer	Peer-to-Peer or SFU communication model
State Management	Real-time synchronization of participant states
Design System	Inter / Manrope / DM Sans typography

Architecture Flow

The system follows a direct media synchronization lifecycle:

User → ConnectMeet UI → Meeting Room → Real-Time Media Layer → Participants

05. CORE VIDEO CONFERENCING WORKFLOW

The workflow facilitates seamless transitions from initial room setup to live collaboration and graceful exit.

Meeting Lifecycle

1. **Dashboard Entry:** High-level selection of New Meeting, Join, or Schedule.
2. **Pre-Join Lobby:** Live camera preview and media hardware configuration.
3. **Live Meeting Room:** Active engagement via audio, video, and collaboration tools.
4. **Media Controls:** Toggling states and switching layouts between Gallery and Speaker views.
5. **Session Clean-up:** Graceful exit with media track disposal for privacy.



06. FEATURE MAP

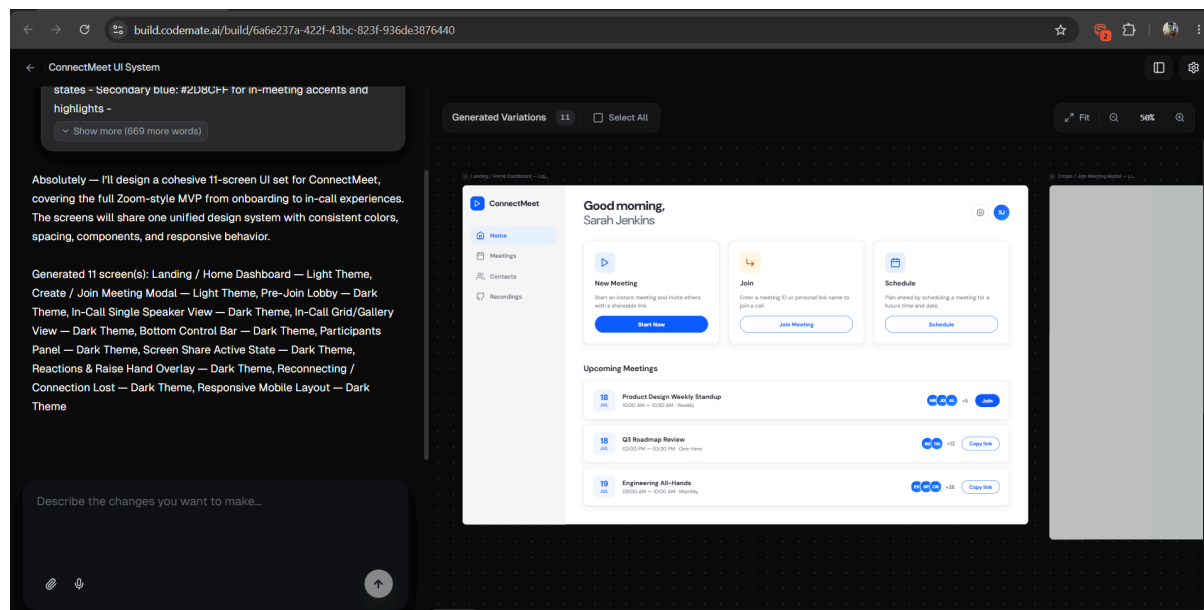
Feature	Description
Pre-Join Lobby	Media previews and device permission state management before entry.
Gallery & Speaker Views	Responsive video grids (up to 16 tiles) or large focused speaker views.
Screen Sharing	Primary canvas promotion for shared content with thumbnail participant strips.
Participant Panel	Live lists with status icons, pinning, and host management tools.
Active Speaker	Audio-level based detection with debouncing to prevent flickering.
Collaboration	Chat, animated emoji reactions, and a chronological raise-hand queue.

07. BUILD: DESIGN MODE

Build Design Mode established a **connected design system** for ConnectMeet, defining visual standards for all 11 screens as a cohesive product.

Visual Standards

The system utilizes an 8px grid with rounded 8–12px corners and pill-shaped buttons. Color tokens prioritize a dark theme for meetings (#0E0E0E) and a light theme for dashboards.

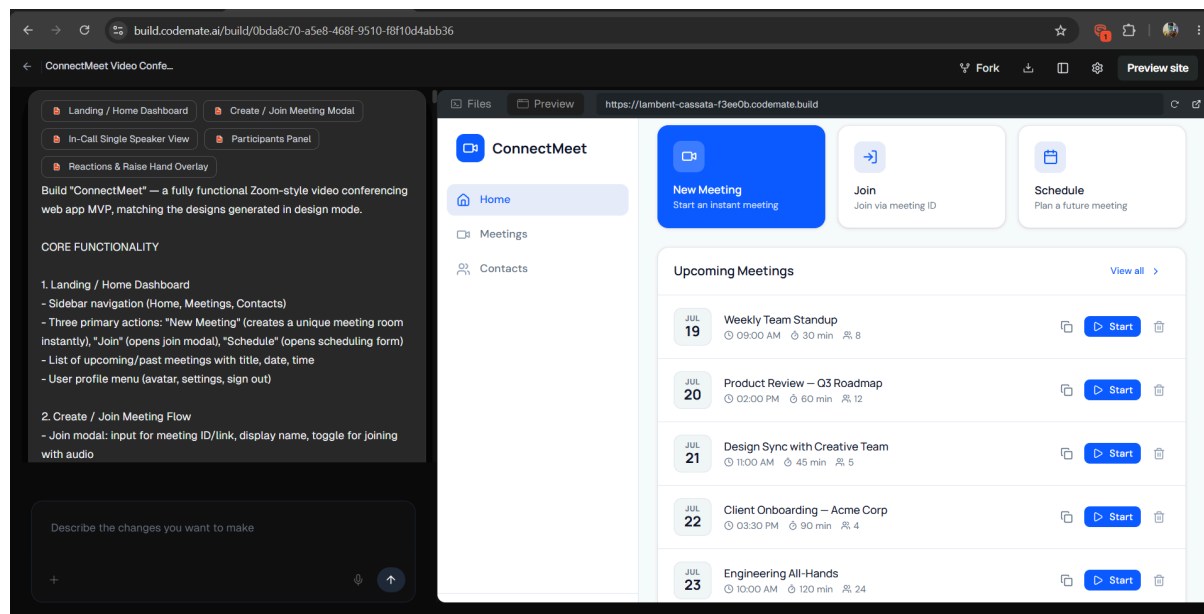


08. BUILD: PROTOTYPE MODE

Prototype Mode converted designs into a **functional conferencing MVP**, implementing core real-time media and participant synchronization.

Functional Core

- **Meeting Creation:** Generates unique rooms and shareable links instantly.
- **Real-Time Sync:** Synchronizes mute, camera, and raise-hand states across all clients.
- **Responsive UI:** Automatically reflows video grids based on screen size and participant count.



09. BUILD PROMPT HIGHLIGHTS

Design Mode

Prompt 1 : UI Design

Design a complete UI screen set for "ConnectMeet" - a Zoom-style video conferencing web app MVP. Generate all screens as a cohesive design system using a professional blue-based visual identity, dark in-meeting interface, light dashboard interface, rounded components, pill-shaped buttons, clear active states and responsive layouts. Include dashboard, meeting creation, joining, pre-join lobby, speaker view, gallery view, participant panel, screen sharing, reactions, reconnecting state and mobile layouts.

Prompt 2 : Connected Design System

Generate all 11 screens as a connected design system sharing the same color tokens, typography scale, spacing and component styles so they feel like one cohesive product.

Prototype Mode

Prompt 3 : Functional ConnectMeet Application

Build "ConnectMeet" - a fully functional Zoom-style video conferencing web app MVP, matching the designs generated in Design Mode. The prompt then defined the dashboard, meeting creation, joining, pre-join lobby, real-time video/audio, participant controls, screen sharing, active speaker detection, pinning, reactions, auto reconnect and responsive behavior.

Prompt 4 : Real-Time Meeting Functionality

Implement real-time peer-to-peer or SFU-based video and audio between meeting participants. Synchronize microphone, camera, hand-raise, pin and active-speaker states across clients while gracefully handling camera failures, permissions, participant joins, host changes and network interruptions.

Prompt 5 : Responsive Meeting Experience

Make the complete application responsive across desktop, tablet and mobile.

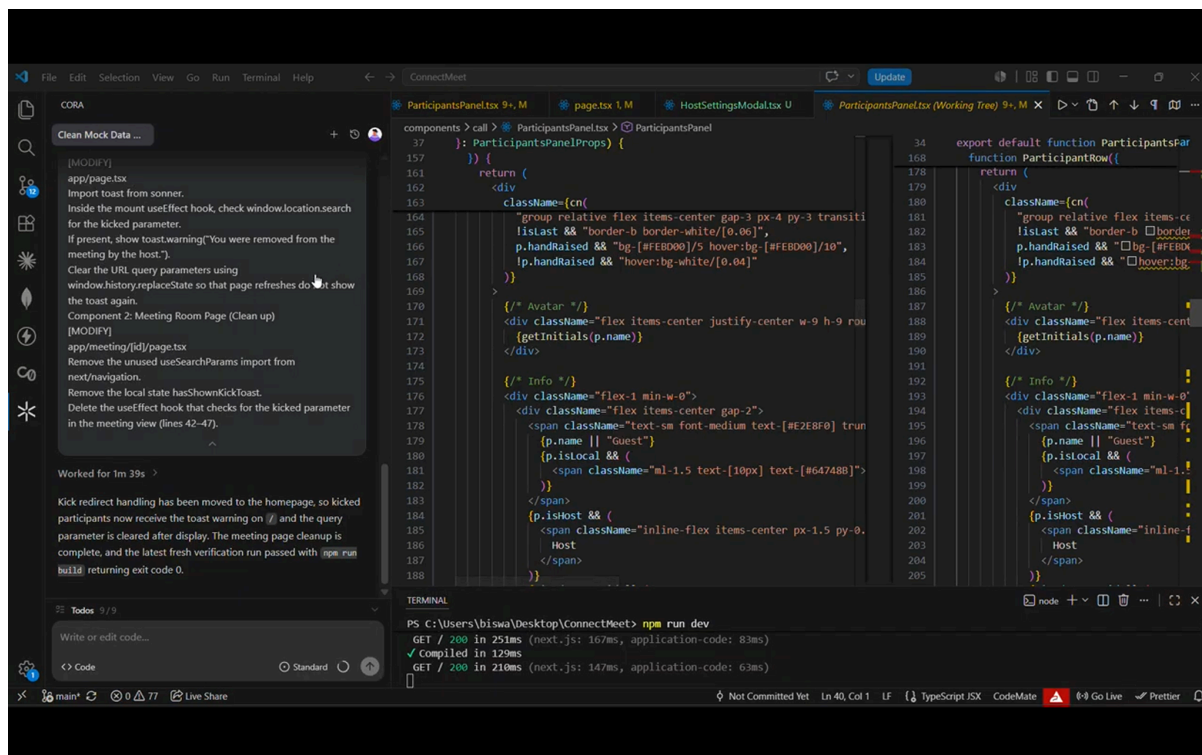
Collapse the bottom control bar into icon-only controls on smaller screens, convert the participant panel into a mobile slide-up sheet and dynamically reflow the video grid based on screen size and participant count.

10. CORA: PLANNING & IMPLEMENTATION

CORA served as the primary development layer for transferring the Build prototype into a refined, functional MVP.

Development Flow

Design Mode → Build Code Mode → Functional MVP → CORA Handoff



11. CORA TASK ARCHIVE

Category	Status Focus
Project Handoff	Exported ZIP from Build for use within CORA development.
Documentation	Requirement specs and design system tokens finalized for implementation.
Infrastructure	WebRTC implementation and backend prompts excluded from initial handoff archive.

12. IMPLEMENTATION MILESTONES

Phase	Milestone	Focus Area
1	Design System	Color tokens, typography scale, and 8px grid system.
2	Foundations	Meeting creation and shareable ID/link validation flow.
3	Core Streaming	Real-time audio/video tracks and pre-join lobby previews.
4	Collaboration	Screen sharing, reactions, and active speaker highlighting.
5	Reliability	Graceful network drop handling and auto-reconnection logic.
6	Responsive UI	Layout reflow for desktop, tablet, and mobile orientations.

13. BACKEND, DATA & REAL-TIME COMMUNICATION

ConnectMeet leverages synchronized media tracks and real-time state management to ensure all participants remain aligned mid-call.

Communication Model

- **Real-Time Media:** Supports synchronized video/audio tracks and participant presence.
- **Architecture:** Allows for either Peer-to-Peer or SFU-based communication for room scalability.
- **State Sync:** Maintains consistent visibility for mute states, hand-raises, and active screen shares across all clients.

Resilience & Edge Cases

The system handles critical interruptions including camera permission denials, network drops, and host departures while preserving meeting state for active participants.

14. AUTHENTICATION & USER MANAGEMENT

User identity is established during the entry flow, collecting a display name that can be edited before joining.

User Features

- **Host Management:** Ability to mute individuals, remove participants, or end the session for all.
- **Profile Management:** User avatars and settings for persistent display identity.
- **Permissions:** Real-time handling of microphone and camera access requests.

Participant Controls

- Mute / Unmute
- Camera On / Off
- Share Screen
- Leave Meeting properly

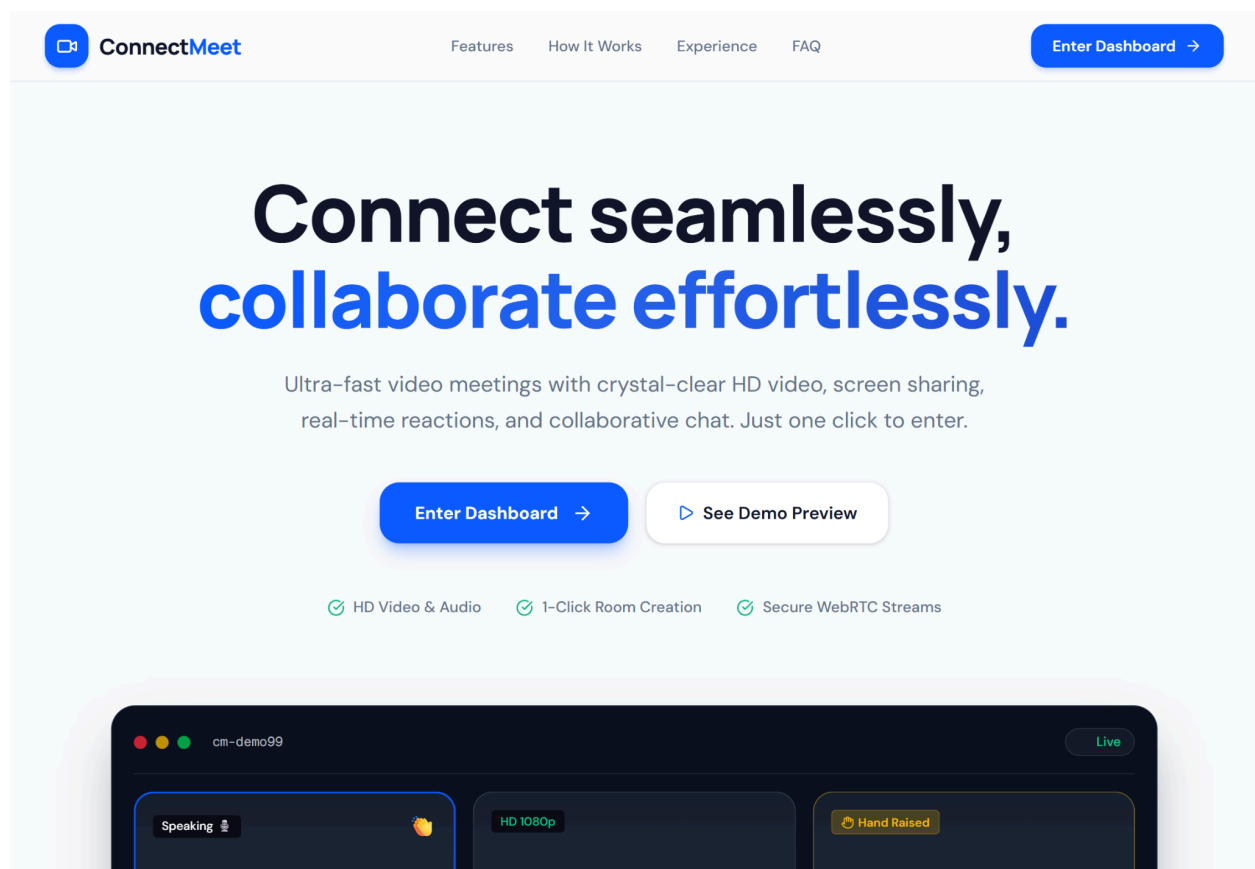
15. RESPONSIVE UI & POLISH

ConnectMeet is built with a **Responsive First** principle, ensuring the video grid and controls adapt naturally across devices.

Detailed visual layouts for desktop, tablet, and mobile platforms are provided below.

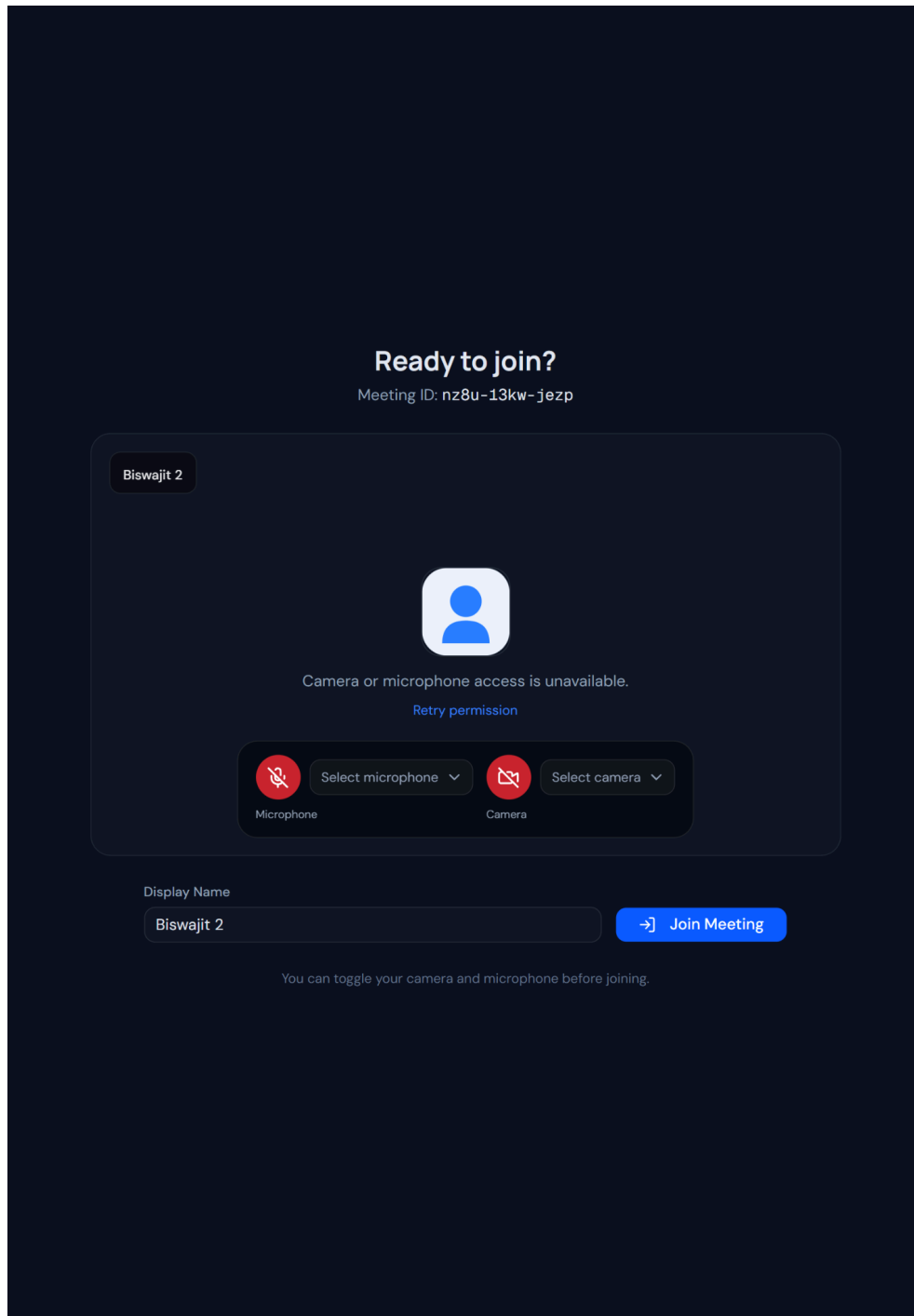
15.1 Desktop View

Features a full meeting control bar and large video canvas supporting complex gallery layouts and side panels.



15.2 Tablet View

Utilizes condensed controls and optimized content density for touch-based navigation.



15.3 Mobile View

Control bars collapse into icon-only buttons, and the participant panel becomes a full-screen slide-up sheet.

